# Anastasios Antoniadis

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## Technical Skills

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|---|---|
| **Languages** | Java, Python, C++, C, Groovy, PHP, Javascript, Datalog, Scheme |
| **Frameworks** | Spring, Protobuff, jQuery, Bootstrap, JSP & Java Servlets |
| **Databases** | Oracle SQL, MySQL, Elasticsearch |
| **Build Tools** | Make, Gradle, Maven, Ant |

## Experience

**University of Athens** Oct 2014 – Dec 2020, May 2024 – Present
*Software Engineer & Research Assistant, PLAST* *Athens, Greece*

- Contributed to the development of static analyses tools, including: the parallelization of data extraction from Java bytecode, a web application to view static analysis results on the source code of the program, and the porting of group's Java bytecode analysis framework to a new Datalog engine.
- Explored ways to improve the scalability and completeness of points-to analysis of Java Enterprise applications e.g., by modeling data structures inside the JDK and web frameworks (Spring, Hibernate, etc.).

*Technologies:* Java, Groovy, Python, LogicBlox, Soufflé, Git.

**Satways Ltd** Mar 2023 – Oct 2023
*Software Engineer* *Athens, Greece*

- Worked on the introduction of data from underwater optic fibers into the company's desktop application, ENGAGE, built on Eclipse RCP.
- Explored the use of the Elastic stack for logging (ELK + Filebeat) and the use of elasticsearch for big data storage.

*Technologies:* Eclipse RCP, Docker, Docker Compose, Elastic Stack, Apache Nifi.

**Cyvox OÜ** Jan 2021 – Jan 2023
*Copywriter, SEO, Developer, and Blog Manager* *Tallinn, Estonia*

- Started a blogging small business and created BORDERPOLAR, a tech and gaming website that reached hundreds of thousands monthly visitors.
- Managed all the website's functional aspects, including content creation (copywriting), SEO, WordPress Development, and Performance Optimization.
- Developed word searching and pattern matching Javascript (SPA) tools for puzzles like Wordle.

*Technologies:* WordPress, PHP, Javascript.

**Oracle Labs Australia** Feb 2019 – Jul 2019
*Research Intern* *Brisbane, Australia*

- Worked on enhancing the completeness of static analysis of Java Enterprise applications using heap snapshots. Experimented with tools for capturing heap snapshots and stack traces and different modes of execution for web apps and frameworks.

*Technologies:* Java, Groovy, Python, Soufflé, Git.

**CERN** Oct 2012 – Oct 2013
*Technical Student, PH-CMG-PS* *Geneva, Switzerland*

- Contributed to the active development of a physics analysis framework, in particular to the production of datasets and management of the group's database.
- Developed a web page for particle collision histogram comparisons, contributed to the creation of C++ libraries for physics using GNU Autotools and the group's migration from CVS to Git.

*Technologies:* Python, Oracle SQL, GNU Autotools, HTML, CSS, PHP, CVS, Git.

## Education

**Doctor of Philosophy – PhD,** Static Program Analysis — 2016 – On Hold
University of Athens, Department of Informatics & Telecommunications
Advisor: Prof. Yannis Smaragdakis

**Master of Science,** Computing Systems — 2015 – 2016
University of Athens, Department of Informatics & Telecommunications
Thesis: "Combining source code metadata and static analysis results via a compiler plug-in"
Grade: 10/10. Valedictorian of graduating class.

**Bachelor of Science,** Computer Science — 2006 – 2014
University of Athens, Department of Informatics & Telecommunications
Thesis: "Declarative Points-to Analysis on Different Datalog Engines"
Grade: 8.51/10. Salutatorian of graduating class.

## Languages

*Greek* (native), *English* (fluent).

## Publications

Anastasios Antoniadis et al. "Universal Scalability in Declarative Program Analysis (with Choice-Based Combination Pruning)". In: *Proc. ACM Program. Lang.* 9.OOPSLA2 (Oct. 2025). DOI: 10.1145/3763129. URL: https://doi.org/10.1145/3763129

Anastasios Antoniadis et al. "Static analysis of Java enterprise applications: frameworks and caches, the elephants in the room". In: *Proceedings of the 41st ACM SIGPLAN Conference on Programming Language Design and Implementation*. PLDI 2020. London, UK: Association for Computing Machinery, 2020, 794–807. ISBN: 9781450376136. DOI: 10.1145/3385412.3386026. URL: https://doi.org/10.1145/3385412.3386026

Sifis Lagouvardos et al. "Static Analysis of Shape in TensorFlow Programs". In: *34th European Conference on Object-Oriented Programming (ECOOP 2020)*. Ed. by Robert Hirschfeld and Tobias Pape. Vol. 166. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2020, 15:1–15:29. ISBN: 978-3-95977-154-2. DOI: 10.4230/LIPIcs.ECOOP.2020.15. URL: https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.ECOOP.2020.15

Tony Antoniadis, Konstantinos Triantafyllou, and Yannis Smaragdakis. "Porting doop to Soufflé: a tale of inter-engine portability for Datalog-based analyses". In: *Proceedings of the 6th ACM SIGPLAN International Workshop on State Of the Art in Program Analysis*. SOAP 2017. Barcelona, Spain: Association for Computing Machinery, 2017, 25–30. ISBN: 9781450350723. DOI: 10.1145/3088515.3088522. URL: https://doi.org/10.1145/3088515.3088522

## Activities

**University of Athens** — Spring 2014 – Fall 2020
*Teaching Assistant*

- **Courses:** Operating Systems, System Programming, Compilers, Systems/Advanced Programming Techniques.
- **Instructors:** Alex Delis, Yannis Smaragdakis

## Army Duties

Fulfilled army duties.

## Grants and Awards

Oracle scholarship for SOAP 2017.

## Hobbies & Interests

Photography, Reading, Films, TV Shows, NFL.